Read the Room: Grasp-Aware Planning and Learned Control for Multi-Robot Transport

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***Abstract*— Shared policy weights do not determine how robots coordinate through a common load. We study this information boundary using grasp-aware planning and learned local control for two wheeled bimanual carriers. An assembly aperture model restricts a recorded beam's admissible yaw from 25.8° (object only) to 5.5°. Aligning global/local geometry and margins recovers a paired eight-scene test from 36/40 to 40/40 without changing the actor. A separate frozen ten-scene test, with randomized geometry, 1–2 kg loads and activation delays up to 8 s, completes 35/40 missions; all failures precede verified-route execution. Local actors share references, while a supervisor selects lift and transport phases. This simulation baseline uses contact-gated welded retention. It motivates a recurrent extension for studying communication amount, message content and task-level agreement; minimum communication and learned readiness remain untested.**

I. Introduction

Scaling data and model capacity offers a route toward a single robot that can perform many tasks. Can the same principle produce a collaborating team? If copies of one policy run on different robots, will their actions naturally complement one another, or must they communicate what they intend to do? Shared weights provide a common action rule, yet the robots have different observations and physical states. They must still agree on intent, coordinate timing and respond to one another.

Cooperative transport makes this problem tangible. A couch may require two robots to reach separated grasps and distribute support. They must acquire contacts and negotiate a route while keeping their local actions compatible with one shared object's motion.

Self-driving offers a useful comparison. Decentralized driving policies can select actions for individual vehicles while accounting for other road users [1]. Their decisions interact, but ordinarily their bodies are mechanically separate. Carriers grasping the same object instead share a dynamical system: one robot's acceleration changes the other's load, and one robot's lift moves the other's grasp target. Decentralizing their controllers therefore requires agreement about both intended motion and physical support as contacts and load distribution change.

Human collaboration suggests a possible interface. Partners can communicate a destination or an instruction such as "turn through the doorway" while adjusting their own movements through vision and touch. Haptic feedback can improve human coordination [2]; high-level conversation does not replace continuous physical feedback. For robots, this motivates shared task intent with local control: communicate goals, grasp roles and motion constraints, then let each robot determine its base, arm and lift actions. Whether those messages suffice when a partner arrives late, slips or changes support is the central challenge.

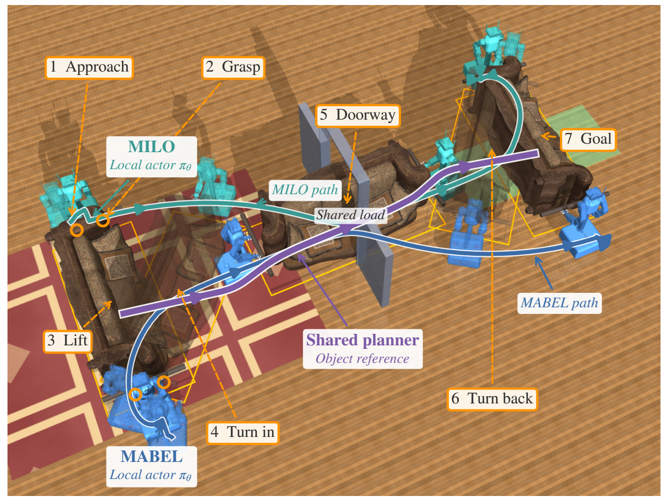

Fig. 1. Recorded cooperative transport (seed 142000). Both robots execute the same local policy with references from one shared high-level planner; a supervisor selects phase and height. Seven numbered stages show approach, grasp, lift and doorway reorientation. Blue/teal curves are measured MABEL/MILO base paths; purple is the planned object path. Curves are projected onto the floor and overlaid for visibility. Translucent poses expose intermediate motion; orange rings mark grasp locations and cutaway walls show the doorway.

We investigate the information boundary of a mechanically coupled team: how much must be shared beyond what each robot can infer through the object, and which information must be explicit? Learned carrying and team policies already demonstrate local cooperation [3]–[5]. Our grasp-aware planning and learned-control framework specifies shared geometry, object references and supervised progression across acquisition and transport. It provides an evaluated baseline for testing whether task-level agreement can replace frequent low-level coordination.

Our contributions are:

- **A grasp-conditioned planning formulation** that incorporates payload and carrier footprints, derives aperture constraints, and maintains consistent geometry between global search and object-level MPC.
- **A continuous acquisition-to-transport framework** combining shared learned local control, independent robot starts and explicit supervised progression without resetting the physical state at handoff.
- **An evaluation of cooperative transport** using paired planning comparisons, recorded asynchronous acquisition and 40 frozen missions, retaining failures and

deviations from planned clearance.

The evaluated stack uses supervised transitions and contact-gated welded retention in simulation. We separately formulate recurrent coordination through object feedback as an extension; reduced communication and hardware performance are not established by these experiments.

II. Related Work

Learning one policy for several robots brings together two lines of work: broadening the skills available to each embodiment and learning how those skills interact during a shared task. TidyBot++ supports the first through a holonomic mobile platform and accessible demonstration collection [6]. CrossFormer brings heterogeneous observations, actions and control rates into one policy [7]. CHORUS connects shared-policy learning to concurrent collaboration: each robot runs the same pretrained VLA, adapted on multi-robot demonstrations, with its own visual observations and an identity/role prompt [5]. Its real-world lifting and transport experiments make teammate reactivity part of the policy's behavior. Robots execute action chunks independently, with a common horizon across control rates. Coordination relies on observing teammates rather than exchanging their states; large timing discrepancies can still move execution outside the demonstration distribution. These results make local observability central to understanding what shared weights provide during collaboration.

Oregon State's cooperative RL work develops the same theme through physical coupling. In decMBC, a geometry-conditioned decentralized controller supports varying arrangements of rigidly attached carriers, including real-world teams of two and three Cassies [3]. decPLM replaces the attachment with sustained surface contact: a learned arm/base hierarchy uses constellation rewards to preserve contact position and orientation [4]. Its task interface supplies feasible contact frames, a synchronized start and contact-frame motion commands; local control maintains support during transport. The comparison between initial-only and continuous contact-frame observations is particularly relevant to information requirements. Initial geometry can support carrying, while continuous pose feedback becomes more valuable for heavier or irregular payloads. These results suggest that useful information depends on the physical demands of the load, not simply on whether execution is labeled centralized or decentralized.

Physical interaction also provides evidence about a partner's behavior. PAINT uses proprioceptive history to infer partner intent during cooperative transport [8], while role-based learning treats a teammate's actions as implicit communication [9]. The training partner matters in this setting. Sequential Asymmetric Imitation first learns with compliant human support, then trains the second robot against the deployed first policy, and finally refines coordination through targeted interventions [10]. Waiting and yielding are learned under the partner behavior encountered during execution. Together, these studies connect the content of local observations to the interaction experience needed to interpret them.

Explicit communication can instead expose information that would otherwise need to be inferred. Distributed MPC exchanges predictions to coordinate coupled motion [11]; learned LSTM encodings compress those messages while retaining trajectory reconstruction at the receiver [12]. Reducing message size thus preserves a different interface from sharing only a task instruction. At the planning level, feasible paired grasps [13] and centralized object paths with local redundancy resolution [14] provide common geometric references for individual controllers. Across these approaches, shared intent, observed object motion and transmitted predictions play distinct roles. Their relationship frames the communication questions studied here.

III. Grasp-Aware Planning

The task specifies an object, assigned hand–rail contacts, a static map and a goal pose. Two wheeled bimanual robots acquire and lift the object before planning its transport. The attained carrier layout defines a planar assembly model for global search; object-level MPC supplies a common motion reference to the local actors (Fig. 2a).

A. Grasp geometry as a team constraint

Let $q = (t, \theta) \in \mathrm{SE}(2)$, $t = (x, y)$, denote object pose and $s_i = (u_i, v_i)$ the nominal position of base i in the object frame. With payload rectangle $\mathcal{P}$ of dimensions (L, D) and base-disc radii r_i,

$$\mathcal{C}(s) = \mathcal{P} \cup \bigcup_{i=1}^{2} B(s_i, r_i), \qquad \mathcal{C}_q = t + R(\theta)\mathcal{C}(s). \tag{1}$$

Calibrated object profiles specify reachable grasps. Base offsets s_i are measured after lift and held fixed for route planning. This is a formation approximation: articulated arms allow the physical bases to move relative to the object. For obstacles $\mathcal{O}$ and signed clearance d, the configuration-space problem [15] is

$$\begin{aligned} &\gamma : [0,1] \to \mathrm{SE}(2), \quad \gamma(0) = q_{\text{lift}}, \quad \gamma(1) = q_g, \\ &d\big(\mathcal{C}_{\gamma(\tau)}, \mathcal{O}\big) \geq \varepsilon \quad \text{for all } \tau \in [0,1]. \end{aligned} \tag{2}$$

The collision model excludes supports below a specified carrying-height bound and represents the payload and bases, not the articulated swept volume.

For a centered opening of width W, define $w_{\mathcal{P}} = L|\sin\theta| + D|\cos\theta|$ and

$$\begin{aligned} w_i(\theta; s) &= 2\big(|u_i \sin\theta + v_i \cos\theta| + r_i\big), \\ \bar{w}(\theta; s) &= \max\{w_{\mathcal{P}}, w_1(\theta; s), w_2(\theta; s)\}. \end{aligned} \tag{3}$$

The aperture condition $\bar{w} \leq W - 2\varepsilon$ is exact for mirrored, equal-radius layouts and conservative otherwise. Yaw scanning and boundary bisection identify admissible intervals.

The corresponding gate-pose set couples orientation and lateral placement:

$$\mathcal{G} = \left\{ (x_d, y, \theta) : |y - y_d| \leq \frac{W - \bar{w}(\theta; s)}{2} - \varepsilon \right\}. \tag{4}$$

Here (x_d, y_d) is the opening center; a negative bound excludes that yaw. This sampling constraint does not certify clearance elsewhere. Figure 3 shows the carriers restricting beam passage.

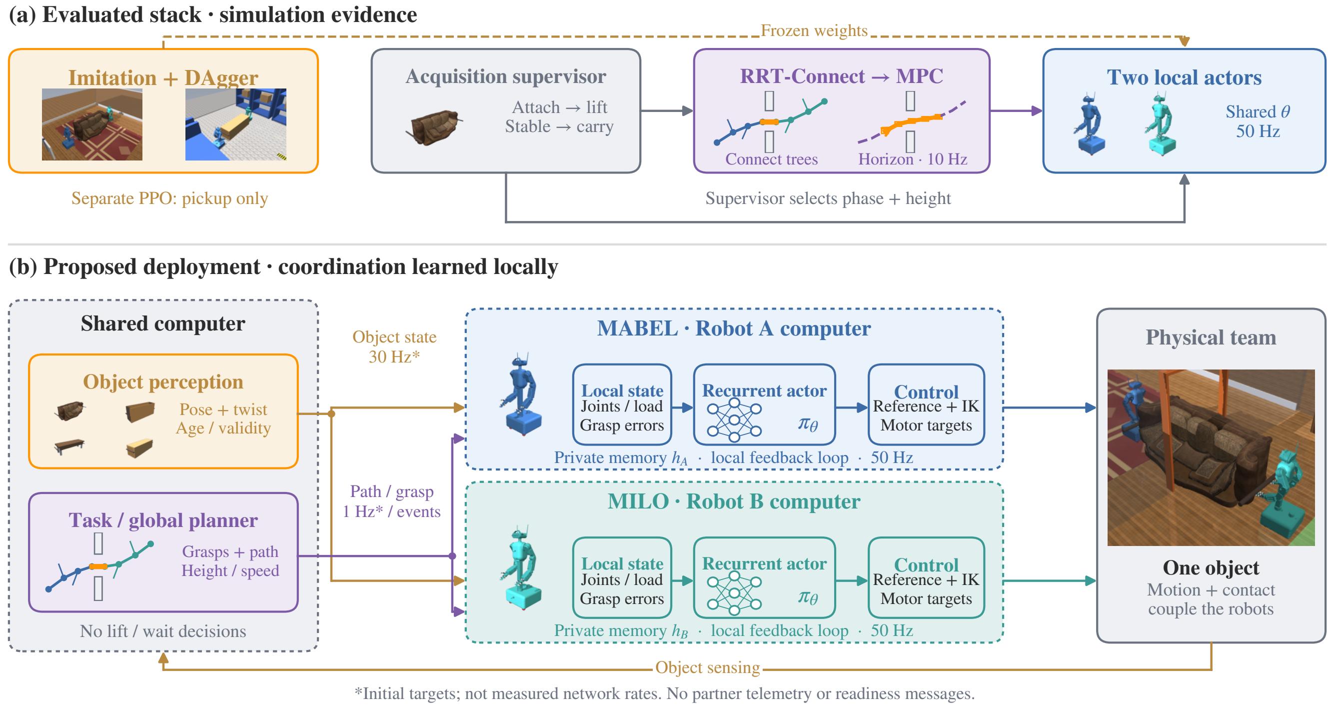

Fig. 2. Shared and local computation. (a) Evaluated imitation actor with supervised transitions; PPO is a separate pickup experiment. (b) Proposed shared perception/planning and robot-local recurrent control. Arrows show data flow; dashed borders delimit computers. Images use recorded models; RRT/MPC sketches are conceptual. Proposed rates are unvalidated targets.

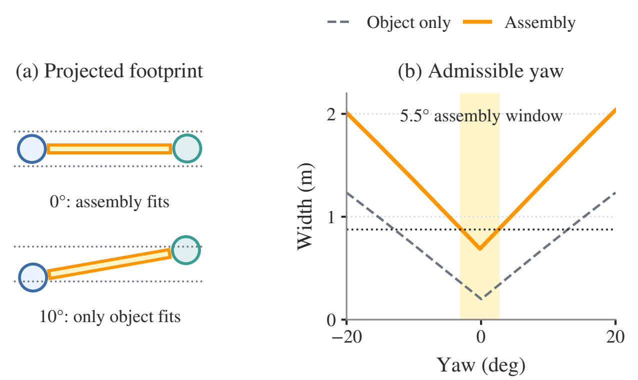

Fig. 3. Why the grasp changes passage feasibility. Analytic projections use the measured beam handoff geometry (seed 142024). (a) Dashed bounds show the available opening width after the 22 mm search margin on each side; the circles are base footprints. (b) The object-only yaw range is about 25.8°, versus 5.5° for the assembly (shaded); the horizontal dotted line is the door budget.

Gate-biased RRT-Connect [16] samples the opposite root, aperture set and workspace with probabilities $(0.10, 0.35, 0.55)$. Steering uses $d_q = \|t_a - t_b\|_2 + 0.6|\operatorname{wrap}(\theta_a - \theta_b)|$ and 0.35 m-equivalent extensions. Search permits three seeded attempts of 20,000 iterations, followed by shortcutting and clearance-checked smoothing. Swept collision checks use 5 mm increments and a 22 mm margin; an independent 10 mm-resampling verifier requires $\varepsilon = 20$ mm along the delivered path, including joins. These finite-resolution checks implement Eq. (2); failed verification rejects the route.

B. Shared receding-horizon object reference

Object MPC [17] predicts q_k under object-frame twist $\nu_k = (v_x, v_y, \omega)$ with timestep h:

$$q_{k+1} = f(q_k, \nu_k) = q_k + h\ \operatorname{diag}(R(\theta_k), 1)\nu_k. \tag{5}$$

At each solve, measured object pose and base offsets update the model. Offsets are parameters, not control decisions. Monotone measured path progress selects $\bar{q}_{1:N}$, so elapsed time alone cannot advance the reference during a stall. With $e_k = q_k - \bar{q}_k$ and locally unwrapped yaw, successive convexification solves

$$\begin{aligned}
\min_{q_{1:N},\nu_{0:N-1},\sigma\geq 0}\ & \textstyle\sum_{k=1}^{N-1}\|e_k\|_Q^2 + \|e_N\|_{Q_f}^2 \\
& + \textstyle\sum_{k=0}^{N-1}\|\nu_k\|_{R_\nu}^2 + \lambda\|\sigma\|_1 \\
\text{s.t.}\quad & q_0 = \hat{q}, \quad q_{k+1} = A_k q_k + B_k \nu_k + c_k, \\
& a_{jk}^\top q_k + \sigma_{jk} \geq b_{jk} + \varepsilon, \\
& |\nu_k| \leq \nu_{\max}, \quad |\nu_k - \nu_{k-1}| \leq h\dot{\nu}_{\max}, \\
& \|\nu_{xy,k} + \omega_k J \hat{s}_i\|_\infty \leq 1.2\ \text{m/s}.
\end{aligned} \tag{6}$$

Here J rotates by 90° and $\hat{s}_i$ is the measured base offset. Affine dynamics approximate Eq. (5); separating planes approximate obstacle clearance with individual nonnegative slacks σ_{jk}. The frozen profile uses $N = 20$, $h = 0.1$ s, $Q = \operatorname{diag}(12, 12, 30)$, $Q_f = 40Q$, $R_\nu = \operatorname{diag}(0.6, 0.6, 0.8)$ and $\lambda = 4000$. Two warm-started convexification passes use OSQP [18]. Limits are 0.12 m/s per translation axis and 0.10 rad/s yaw, with slew bounds of 0.15 m/s^2 and 0.10 rad/s^2. The first control is accepted only if the solve succeeds, the command is finite, and maximum slack is at most 10 mm; otherwise the bridge commands zero twist.

Execution clearance is measured independently of the QP slacks.

For world-frame station offset ρ_i from the object center, the common reference induces rigid-motion feedforward

$$v_i^{\text{ff}} = R(\theta)(v_x, v_y)^\top + \omega\hat{z} \times \rho_i, \qquad \omega_i^{\text{ff}} = \omega. \tag{7}$$

Each robot transforms this reference into its base frame and adds bounded learned residuals. Both planners use the same assembly geometry and requested clearance. Reference updates preserve the continuous physical state.

IV. Local Control and Coordination

A. Shared policy and local observations

Each robot executes $a_i = \pi_\theta(o_i)$ at 50 Hz with identical weights; MuJoCo dynamics run at 200 Hz [19]. The 157-dimensional observation includes joint state and tracking errors, base motion and gravity, object-relative pose/twist, grasp geometry, palm-target and docking errors, phase/height references, previous action and commanded motion. It excludes partner joint/velocity streams. Simulation supplies object state and six clipped constraint-multiplier channels for grasp feedback.

For base position p_i and yaw ψ_i, relative features are

$$\begin{aligned} \tilde{p}_i &= R(-\psi_i)(t - p_i), \qquad \tilde{s}_i = R(\theta - \psi_i)s_i, \\ c_i &= \big(\cos 2(\theta - \psi_i),\ \sin 2(\theta - \psi_i)\big). \end{aligned} \tag{8}$$

The double-angle feature represents footprint symmetry; assigned hand targets retain contact identity. This common coordinate convention permits weight sharing between opposed carriers without assuming policy equivariance. The MLP (Table II) produces three base and 18 upper-body actions. Arm targets are posture residuals; lift actions specify a bounded 0.06 m/s rate during lifting and position residuals around the attained carrying posture during transport.

B. Acquisition and asynchronous progression

Assigned rails define the approach targets. The geometric teacher stages, docks and refines wrist pose within 40 mm. Panel/beam profiles use 0.18 m hand spacing instead of 0.40 m and reserve 60 mm downward lift travel. Robot i attaches when both assigned palms are within 35 mm and 30° of their targets, base speed is below 0.10 m/s, and both palm/finger contacts apply more than 0.1 N compression to the correct rails. Weld constraints then retain the attained transforms [20]; subsequent frictional slip is not modeled.

One actor starts immediately and the other after $\Delta \in [0, 8]$ s; the delayed robot holds its initial command until activation. Let b_i indicate attachment and z_0 initial object height. The attached robot's height reference is

$$z_i^* = \begin{cases} z_0 + 0.04\ \text{m}, & b_i = 1,\ b_1 b_2 = 0, \\ z_0 + 0.12\ \text{m}, & b_1 b_2 = 1. \end{cases} \tag{9}$$

During early lift, the base holds and upward action is clipped at a 40 mm centroid rise. Remaining support permits this partial lift while the late robot tracks its moving grasp targets. Joint attachment releases the full lift. Transport begins after height error remains below 20 mm and object roll/pitch below 0.06 rad for 25 control steps (0.5 s). The same actor then tracks MPC references without resetting the physical state. Gated-start comparisons instead synchronize controller availability and attachment.

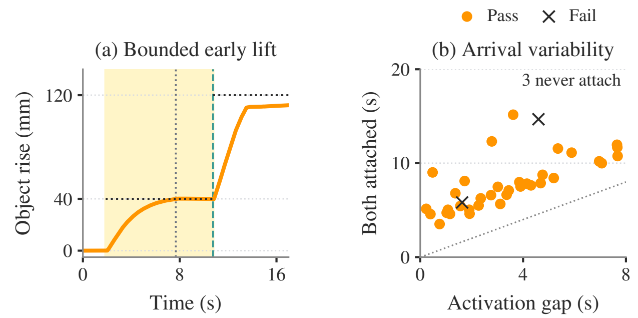

Fig. 4. Independent arrival. (a) Loading dock, seed 142013: measured object rise under a 7.68 s activation gap. Shading marks early lift; vertical dotted/dashed lines mark late activation/second attachment; horizontal lines mark height targets. (b) Both-attachment times for the 37 trials that acquire the object; color marks final mission outcome. The other three trials never attach both robots. The diagonal is $t = \Delta$.

C. Information allocation

The evaluated supervisor reads attachment state and returns phase/height references through shared memory; object MPC updates at 10 Hz. Thus local action inference coexists with shared progression. The proposed deployment (Fig. 2b) transmits grasp geometry, spatial path, carry height and speed limits at setup/replanning events (initial target: 1 Hz), and timestamped object pose, twist and validity at 30 Hz. Robot-local computation retains encoders, hand kinematics, contact/load estimates, recurrent state and 50 Hz motor-reference generation. No partner telemetry or readiness messages are proposed; heartbeats and faults remain system traffic. Central perception still requires object-state transmission. These rates are design targets; network timing and hardware sensing are untested.

D. Proposed coordination through object feedback

The proposed extension places reference following and action inference on each robot. A spatial path indexed by measured object progress supplies shared intent, while a recurrent actor determines whether to lift, hold or move:

$$\begin{aligned} (a_{i,t}, h_{i,t+1}) &= \pi_\theta(o_{i,t}, h_{i,t}), \\ o_{i,t} &= [x_i, e_i, \hat{x}_{O,i}, g_i, \mathcal{R}_i, m_i, a_{i,t-1}]. \end{aligned} \tag{10}$$

Here x_i is proprioception, e_i combines hand-target errors and contact/load feedback, $\hat{x}_{O,i}$ is object pose/twist and support-relative height, g_i specifies grasps, $\mathcal{R}_i$ contains path/height/speed references, and m_i records measurement age/validity. Memory h_i remains private. Features use calibrated base frames and preserve grasp identity.

This actor replaces supervised phase selection and must have authority to stop base feedforward. Object motion can reveal partner effects but cannot uniquely distinguish partner support from environmental support. Training therefore requires contact and load history, physical grasp retention, and variation in arrival, stalls, slip and sensing delay. A privileged critic can shape supported progress and smooth

TABLE I

PER-ROBOT ACQUISITION REWARDS IN THE SEPARATE PPO COMPARISON. RATES ARE MULTIPLIED BY $\Delta t = 0.02$ S; EVENT REWARDS ARE ADDED ONCE. THE FROZEN FULL-SYSTEM ACTOR INSTEAD MINIMIZES IMITATION LOSS.

Term	Rate / event value
Approach progress	$6\,\mathrm{clip}((d_{t-1} - d_t)/\Delta t, -0.1, 0.1)$
Proximity	$2\exp[-(d_t/0.10)^2]$
Alignment, original	$\exp[-(\alpha_t/0.35)^2]$
Alignment, gated	$\exp[-(\alpha_t/0.35)^2 - (d_t/0.15)^2]$
Lift height	$10H(z_t - z^*)$
Lift levelness	$3\exp[-\|\eta_t\|^2/0.15^2]$
Approach / lift costs	$-0.5 - 4(\beta_i/\beta_{\max})^2$
Transition / failure	$+2$ each at both-attached and lift-complete; -5 on physical termination.

motion while penalizing dragging and excessive interaction loads. This recurrent design is prospective; the experiments below evaluate the supervised feedforward baseline.

V. EXPERIMENTAL EVALUATION

A. Research questions and scope

The study is motivated by three questions: **RQ1**, how much information must a mechanically coupled team share beyond local sensing and object feedback? **RQ2**, which information should be communicated—task intent, geometric constraints, object state or partner events? **RQ3**, can task-level agreement support local control under changing contact, load and partner behavior? The present experiments characterize a supervised baseline through geometric compatibility, asynchronous acquisition and mission reliability. Communication-budget ablations and recurrent control remain prospective.

B. Learning and execution protocol

The mission actor is trained by imitation and DAgger [21]. A geometric teacher labels approach/lift, and a velocity actor labels transport. Three aggregation rounds collect 30 episodes each: teacher execution first, then student execution with teacher labels and no interventions. Stable-hold collection succeeds in 86/90 episodes. The final dataset contains 524,490 local observations, including 345,484 imported samples. Phase-balanced weighted action MSE uses the settings in Table II and frozen normalization. The mission actor receives no subsequent PPO update.

Two separate PPO branches each run 65,536 steps with an asymmetric 181-input critic [22], [23]. They use generalized advantages [24], a 0.3 KL anchor, value coefficient 0.5 and zero entropy coefficient. Table I defines their acquisition rewards; the variants differ only in distance gating of orientation reward. Transport rewards retain common command-tracking, posture, grasp and regularization terms, detailed in the accompanying reward specification. All evaluations use deterministic actor means.

For maximum hand-position/orientation errors d_t, α_t, object roll/pitch η_t and base inclination β_i, the normalized

TABLE II

COMPACT LEARNING AND EXECUTION SETTINGS.

Component	Configuration
Actor / critic input	157 / 181 values; 21 actions per robot.
Network	ELU MLP, 512–256–128 hidden units.
Physics / policy	200 / 50 Hz; identical actor weights.
Imitation	Adam 3×10^{-4}; 3 fits, 40 epochs each.
PPO rollout	1 environment, 128 steps; 512 updates.
PPO optimization	3×10^{-5} learning rate; 3 epochs, 4 minibatches; $\gamma = 0.99$, $\lambda = 0.95$.
PPO protection	Clip 0.2, anchor coefficient 0.3; 32 critic-only warmup updates; frozen normalization.

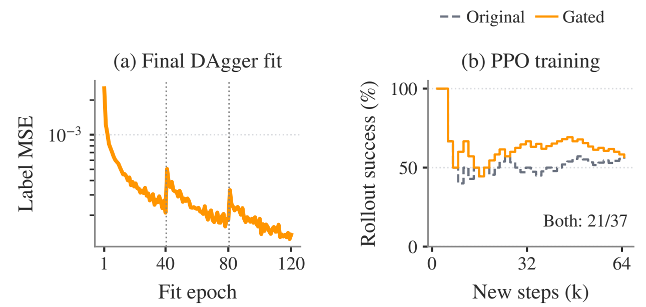

Fig. 5. Training outcomes. (a) Weighted imitation loss across three 40-epoch fits; dotted lines mark dataset aggregation. Collection succeeds in 30/30 teacher episodes, then 28/30 and 28/30 student episodes. (b) Cumulative success over completed PPO training episodes; both reward variants finish at 21/37 after 65,536 new steps.

height term is

$$\Phi(e) = \left(e^{-(e/0.08)^2} + 0.5e^{-(e/0.25)^2}\right)/1.5,$$
$$H(e) = \max\left\{0, \frac{\Phi(e) - \Phi(e_0)}{1 - \Phi(e_0)}\right\}, \tag{11}$$

where $e = z - z^*$ and $e_0 = z_{\mathrm{rest}} - z^*$. At zero initial error, $H = \Phi$. The inclination scale is $\beta_{\max} = 2°$.

DAgger collection randomizes mass $U[1, 2]$ kg, dimension factors $U[0.95, 1.05]$, rail friction $U[0.9, 1.8]$, base offsets 0.08–0.50 m, headings $\pm 20°$ and activation delays 0–8 s across ten scenes. The eight-scene PPO branches use mass $U[0.75, 3]$ kg, dimension factors $U[0.9, 1.1]$, offsets 0.08–0.65 m, headings $\pm 30°$ and delays 0–2 s, with COM perturbations $\pm(0.05, 0.12, 0.05)$ m and servo-gain factors 0.85–1.15. Automatic stage/difficulty curricula are disabled; DAgger changes the executing teacher/student across rounds. Training ends at 35 s by truncation or earlier upon physical failure; successful DAgger collections also stop early. Figure 5 reports optimization and stochastic collection outcomes, distinct from the frozen evaluations.

C. Mission test and outcome criteria

The frozen test comprises four prespecified seeds in each of ten scenes (Table III): nine doorway environments and an open arena. Each mission starts detached and has a 180 s simulation budget. Mass, dimensions, base offsets, headings and activation delays use the DAgger ranges above; object XY and yaw additionally vary by ± 0.10 m and ± 0.20 rad. Thus visually large payloads represent 1–2 kg loads. The

Fig. 6. Recorded sofa, wardrobe, table, crate and beam missions (seeds 142000, 142004, 142008, 142016, 142024). Columns show attachment, lift, doorway passage, goal arrival and measured/planned paths. Times are from simulation; Table III includes all ten scenes and failures.

TABLE III

SCENE GEOMETRY AND FROZEN OUTCOMES: FOUR TRIALS PER SCENE.

Scene	Object	$L \times D \times H$ (m)	S (%)	T_L	T_G
Living	Sofa	$2.73 \times 1.12 \times 0.93$	75	9.0	98.3
Bedroom	Wardrobe	$2.10 \times 0.62 \times 0.95$	100	11.7	90.9
Dining	Table	$2.40 \times 0.76 \times 0.75$	100	8.6	94.9
Dock	Case	$2.80 \times 0.82 \times 0.80$	100	9.3	106.3
Warehouse	Crate	$2.20 \times 0.70 \times 0.65$	100	9.7	96.4
Gallery	Panel	$1.90 \times 0.09 \times 1.25$	50	11.6	93.1
Workshop	Beam	$3.20 \times 0.20 \times 0.20$	75	15.0	122.1
Office	Table	$2.40 \times 1.10 \times 0.74$	75	9.1	93.5
Apartment	Sofa	$2.10 \times 0.86 \times 0.78$	100	10.7	91.8
Arena	Sofa	$2.10 \times 0.90 \times 0.83$	100	11.3	40.3
All (40)		–	87.5	10.6	95.4

Nominal dimensions; variation $\pm 5\%$. T_L/T_G: median seconds to completed lift/first goal among passes. Simulation time; planning compute is excluded.

TABLE IV

EVALUATION PROTOCOLS AND ORIGINAL DENOMINATORS.

Test	Domain / comparison	Pass
Frozen missions	10 scenes; 1–2 kg; independent starts, $\Delta \leq 8$ s; 180 s.	35/40
Original MPC	8 scenes; 1.5 kg; gated starts, $\Delta \leq 0.8$ s; 180 s.	36/40
Aligned MPC	Same actor and 40 seeds as original MPC; geometry and margin change.	40/40
PPO original	8 scenes; pickup only; 1.5 kg; gated starts, $\Delta \leq 2$ s; 35 s.	40/40
PPO gated	Same 40 pickup seeds; distance-gated orientation reward.	39/40

paired MPC and pickup-only PPO tests use separate actors and domains (Table IV); their results are not pooled.

Let F denote physical termination at any point in a mission. Failure guards reject object heights outside $[z_{\text{rest}} - 0.12, z_{\text{rest}} + 0.30]$ m, object roll/pitch exceeding 0.6 rad, base inclination above $2°$, or forbidden world contact. Base inclination is checked every physics substep. With final object position t_T, yaw error $\delta\theta_T$, completed-lift flag ℓ, attachment flags b_i, and minimum measured assembly clearance $d_{\min}$, mission success is

$$S = \mathbf{1}\{b_1 b_2 = 1,\ \ell = 1,\ \|t_T - t_g\| < 0.10,\ |\delta\theta_T| < 0.10,\ d_{\min} \geq 0,\ \neg\mathsf{F}\}. \tag{12}$$

Position and angle tolerances are in meters and radians. Incomplete acquisition/lift and rejected routes count as failures. Planned clearance (20 mm) and measured execution clearance (0 mm) are separate criteria. All original trials remain in the denominator; figures replay their recorded states. Computation time is reported separately from simulation duration.

D. Geometric compatibility and asynchronous acquisition

The paired 40-start comparison yields 36/40 successes with the original MPC and 40/40 after aligning its payload geometry and requested margin with the global planner. The original disc cover and 30 mm margin reject motions admitted by the global rectangle model at 20 mm. The joint change recovers four living-room stalls with unchanged actor weights (Fig. 7a). This is a paired integration comparison, not an

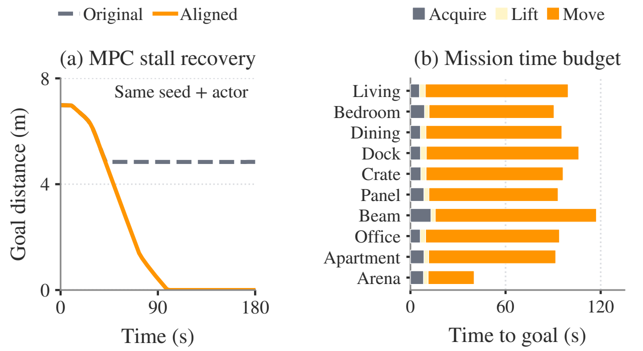

Fig. 7. Planning consistency and mission duration. (a) Same-seed, same-actor MPC comparison (114000): the original model stalls before the doorway; aligned geometry and margin permit progress to the goal. (b) Mean phase durations among successful frozen missions in each scene. Navigation dominates elapsed simulation time. The paired development case and frozen test use different actors and domains.

isolated footprint ablation: geometry and margin change together, and the archived configurations differ beyond a single controlled parameter. The result motivates shared geometric constraints in **RQ2**.

Under independent arrival, the illustrated 7.68 s activation gap produces attachment at 1.88 and 10.76 s and transport at 13.44 s (Fig. 4). The early robot lifts while its partner approaches, but the supervisor still selects joint progression. This successful trace characterizes the baseline for **RQ3**; it does not establish learned readiness or robustness to every delay. The delayed beam failure links timing sensitivity to acquisition geometry.

The separate pickup-only PPO comparison yields 40/40 and 39/40 for the original and distance-gated rewards. Neither includes full navigation or a matched unrefined all-object baseline, so it provides no evidence of improved mission reliability.

E. Reliability across acquisition and transport

The complete frozen test succeeds in **35/40 missions (87.5%)**, below the prespecified $> 90\%$ target. Failures comprise two panel tip-overs before grasp, a delayed beam acquisition timeout, an incomplete office-table lift and an exhausted living-room route search. Of 40 starts, 37 achieve joint attachment, 36 complete lift, and 35 obtain and execute a verified route (Fig. 8a). The conditional 35/35 transport result applies to successfully acquired states; acquisition remains part of team reliability. Four trials per scene support descriptive comparisons rather than precise scene-specific success estimates.

For 31 successful doorway missions, median planned/measured minimum clearances are 32.5/29.0 mm, with a median within-trial difference of 2.2 mm. One reaches 19.95 mm: above the execution threshold but below the planning margin. These measurements concern control-step payload/base footprints, not continuous clearance of the articulated robot geometry. Among successful missions, median first goal arrival is 95.4 s and completed lift 10.6 s; transport accounts for a median 88.6% of duration (Fig. 7b).

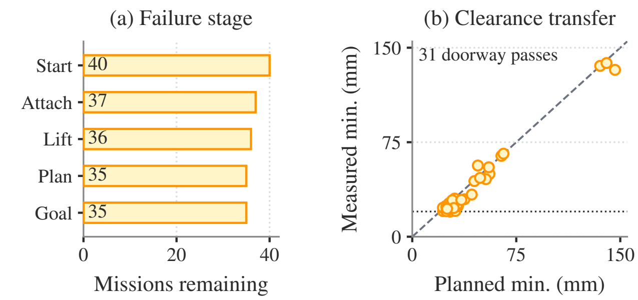

Fig. 8. (a) Mission counts at each stage, retaining all 40 starts. (b) Planned versus measured minimum clearance for 31 successful doorway missions; four arena passes have no finite clearance. Diagonal: equality; horizontal dotted line: 20 mm planning requirement.

Median search and smoothing times are 5.15 and 19.83 s; their per-mission sum has median 26.45 s (11.41–70.70 s), excluding verification and retries. These wall-clock costs are excluded from simulated completion times. Configured 10 Hz MPC and 50 Hz actor rates therefore do not establish real-time execution.

VI. Discussion

The results distinguish local action inference from shared coordination. For **RQ1**, excluding partner telemetry does not eliminate communication: object estimates, geometric references and progression decisions remain shared. A communication minimum is therefore a reliability–bandwidth relationship, with perception and partner traffic accounted for separately. The present activation-delay tests characterize asynchronous task execution, not that relationship.

For **RQ2**, the aperture analysis and paired planning comparison show why a goal alone leaves important constraints unspecified. Contact geometry determines admissible motion, while object state reveals its execution. Prediction exchange, readiness events and local load feedback expose different aspects of the team state; equal message rates need not provide equivalent information. Comparisons at matched budgets are needed to identify useful message content.

For **RQ3**, partial lifting and delayed acquisition coexist with a supervisor that releases joint progression. Replacing this supervisor requires an actor to distinguish adequate support from contact with the environment and to recover from partner stalls or slip. Recurrent state and load feedback are plausible mechanisms, but welded retention does not test these behaviors. The five pre-transport failures also show why evaluation must include acquisition rather than condition success on an established grasp.

VII. Conclusion

We presented grasp-aware planning and shared learned control for cooperative transport, with explicit geometric, observation and progression interfaces. The frozen simulation study completes 35 of 40 missions and identifies acquisition failures and planning-model disagreement as coordination bottlenecks. It establishes a supervised baseline for studying how much information to communicate, which information matters, and when shared task intent can support locally controlled collaboration.

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